base pair substitution mutation and base pair insertion or deletions.

Frame shift mutation usually results in a nonfunctional protein. An addition or deletion of one base pair, for example, shifts the mRNA's downstream reading frame by one base, so that incorrect amino acids are added to the polypeptide chain after the mutation site. Point mutations are single base changes, that do not affect the reading frame, that is, the mutation only makes a single change in a single codon,

Mutations can also be defined according to their effects on amino acid sequences in proteins. They are missense mutation, silent mutation, nonsense mutation. Forward mutations change the genotype from wild type to mutant and reverse mutations (or reversions or back mutations) change the genotype from mutant to wild type or to partially wild type.

The term mutant refers to an organism in which either the base sequence of DNA or the phenotype has been changed. The process of formation of mutant organism is called mutagenesis. Ames test is an indicator of whether the chemical is a mutagen. The Ames test assays the ability of chemicals to revert mutant strains of the bacterium Salmonella typhimurium to wild type. The most commonly employed gene transfer methods are transformation, conjugation, transduction, electroporation, lipofection and direct transfer of DNA. Transformation is genetic alteration of a cell resulting from the direct uptake, incorporation and expression of exogenous genetic material (exogenous DNA) from its surroundings. Competence refers to the state of being able to take up exogenous DNA from the environment. During conjugation, two live bacteria (a donor and a recipient) come together, join by cytoplasmic bridges (e.g. pilus) and transfer single stranded DNA (from donor to recipient). Transduction is the transfer of bacterial genes from one bacteria to other by viruses, e.g. Bacteriophage (Bacterial viruses). **Recombination** is the process in which one or more nucleic acids molecules are rearranged or combined to produce a new nucleotide sequence. Cloning in the molecular biology sense (as opposed to cloning whole organisms) is the making of many copies of a segment of DNA, such as a gene. Cloning makes it possible to generate large amounts of pure DNA, such as genes, which can then be manipulated in various ways

Vectors are the DNA molecules, which carry a foreign DNA fragment to be cloned. They are cloning vehicles, examples of which are Plasmids, Bacteriophages, cosmids, phagemids and artificial chromosomes. Bacterial plasmids are extra chromosomal elements that replicate autonomously in cells. They are viruses that replicate within the bacteria. Cosmids are the vectors possessing the characteristics of both plasmid and bacteriophage. Phagemids are the combination of plasmid and phage, and can function as either plasmid or phage. The plasmid vectors that are specifically designed to replicate in two or more different host organisms (say in E. *coli* and yeast) are referred to as shuttle vectors. An expression vector is a cloning vector containing the regulatory sequences (promoter sequence) necessary to allow the transcription and translation of a cloned gene or genes. Restriction enzymes are the

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